

Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering

System Design & Software Architecture

Academic Performance Monitoring and AI-Based Recommendation System

Course	CSE 4204 — Mobile Computing Lab
Section	8A
Team	T07
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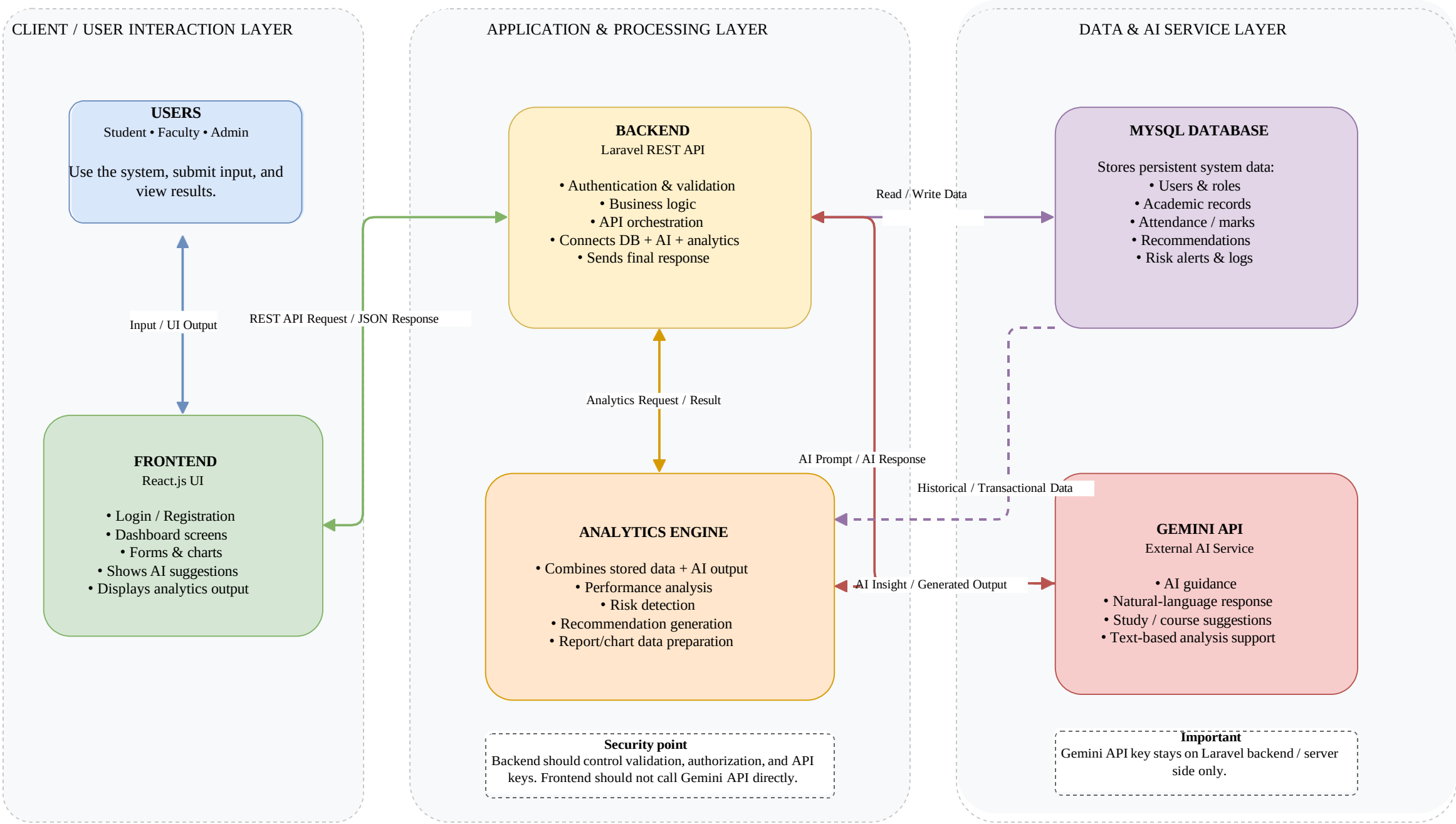
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System Architecture Diagram

React.js · Laravel · MySQL · Gemini API · Analytics Engine

System Architecture Diagram

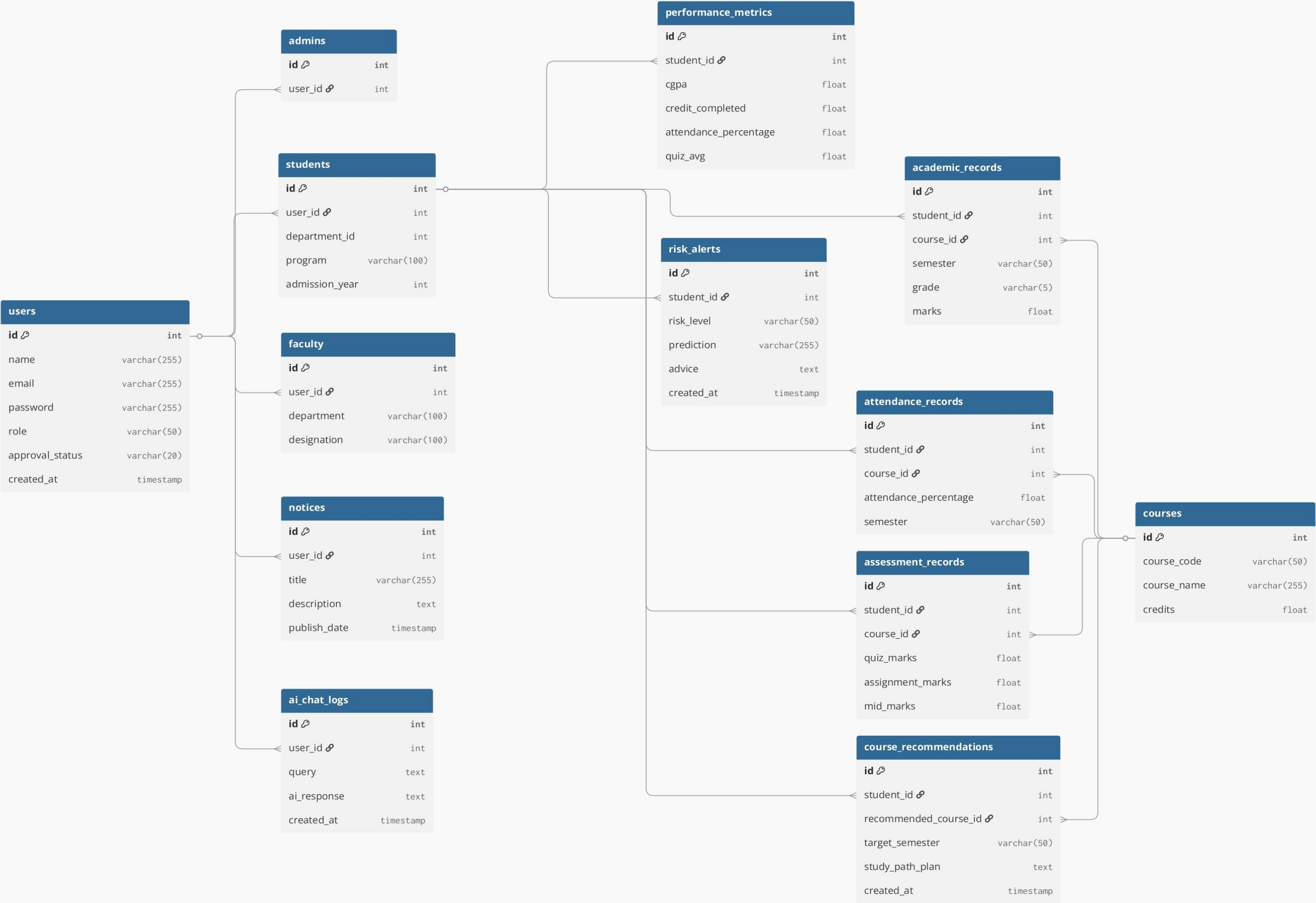
React.js Frontend • Laravel Backend • MySQL • Gemini API • Analytics Engine

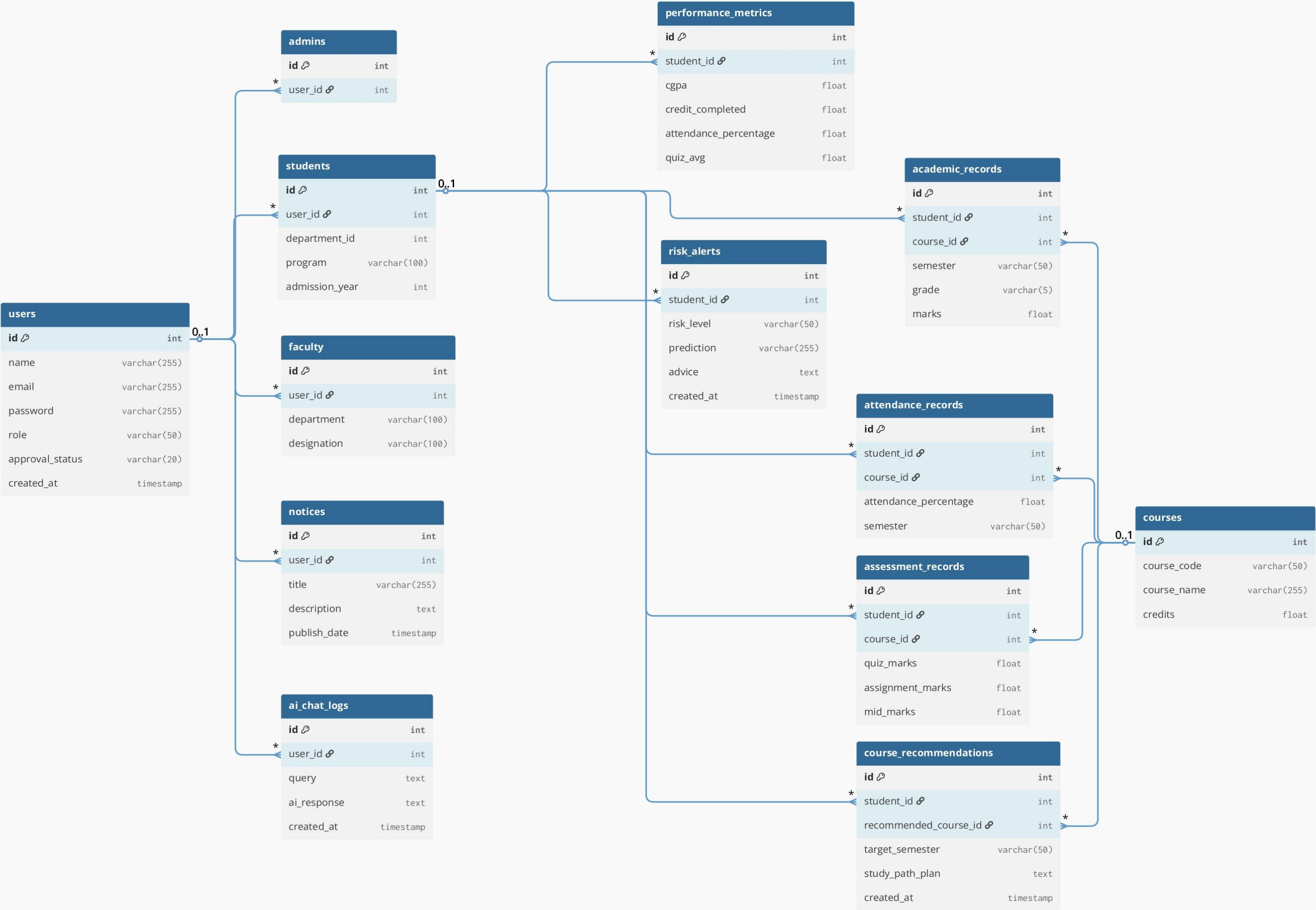


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ER Diagram (Database Design)

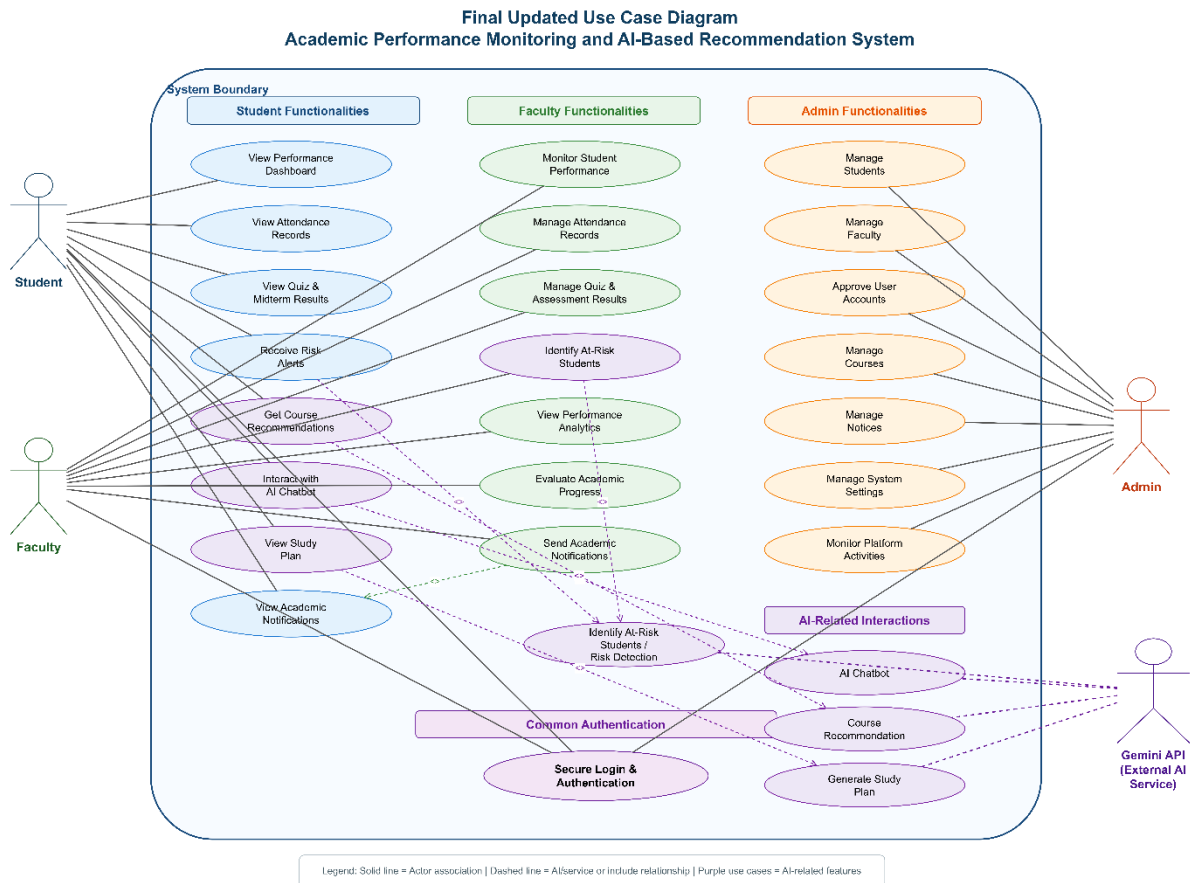
Entity-Relationship Diagram showing all tables, keys, and relationships





3.Updated Use Case Diagram

The Use Case Diagram illustrates the interactions between system actors (Student, Faculty, Admin) and major system functionalities including AI-related interactions.



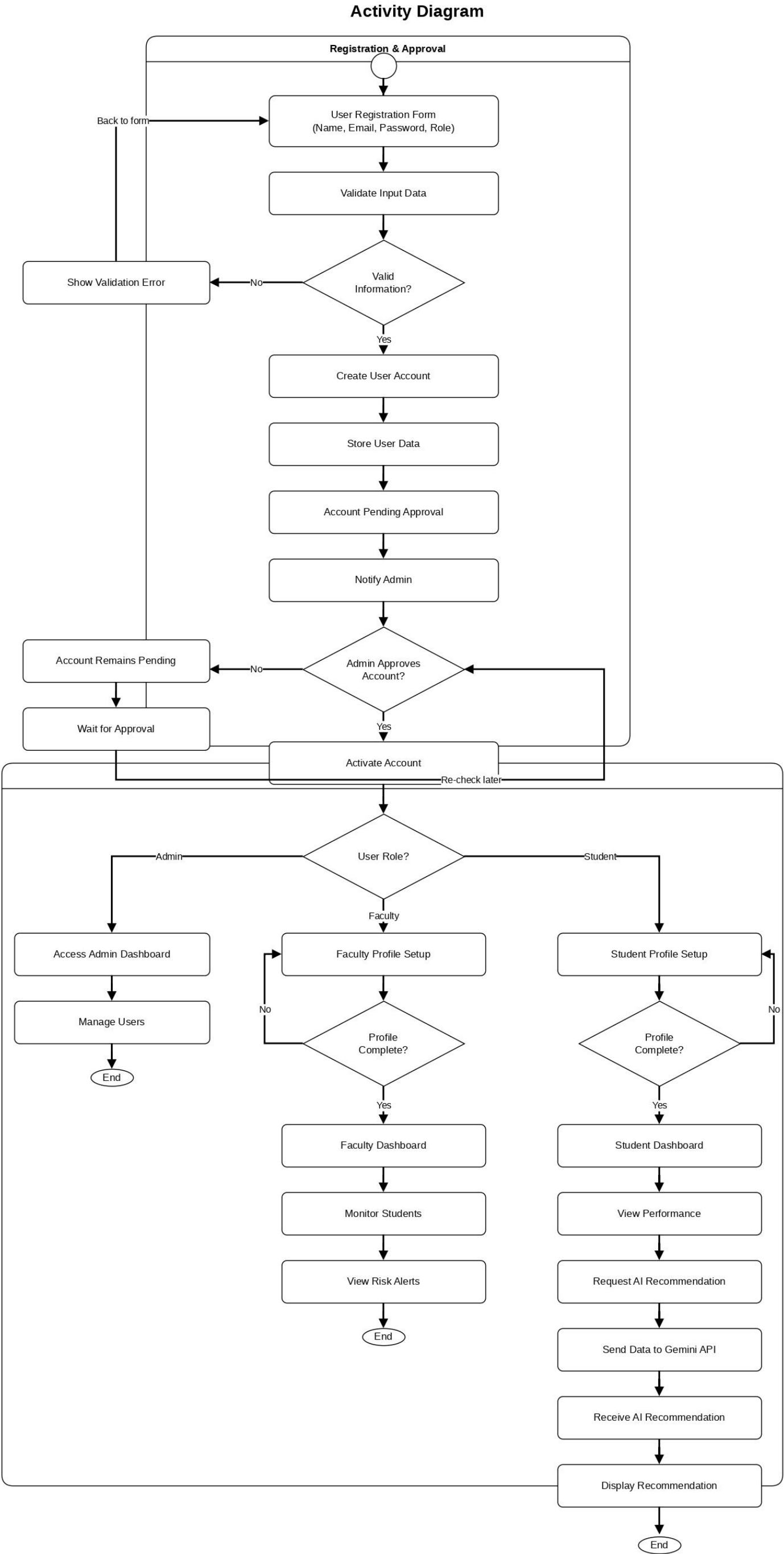
Actors included: Admin • Student / Faculty • Moderator

Major functionalities and AI-related interactions are shown in the diagram above.

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Activity Diagram

Registration & Approval — Role-Based Flow After Account Activation



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API Design Document

Planned API endpoints for Authentication, Student, Faculty, Admin, AI, and Notifications

5. API Design Document

Academic Performance Monitoring and AI-Based Recommendation System

This section presents the planned API endpoints for the proposed system. The APIs are organized by user role and system functionality, including authentication, student activities, faculty operations, admin management, Gemini AI integration, and notification services. Each endpoint includes its purpose, required input data, and expected output.

Base URL	
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API Endpoint Summary

SL	API Name	Method	Endpoint	Category
1	User Registration	POST	/auth/register	Authentication APIs
2	User Login	POST	/auth/login	Authentication APIs
3	User Logout	POST	/auth/logout	Authentication APIs
4	Get User Profile	GET	/profile	Authentication APIs
5	View Dashboard	GET	/student/dashboard	Student APIs
6	View Attendance	GET	/student/attendance	Student APIs
7	View Academic Results	GET	/student/results	Student APIs
8	View Risk Alerts	GET	/student/alerts	Student APIs
9	Monitor Student Performance	GET	/faculty/students/performance	Faculty APIs
10	Manage Attendance	POST	/faculty/attendance	Faculty APIs
11	Manage Assessment Results	POST	/faculty/results	Faculty APIs
12	View At-Risk Students	GET	/faculty/risk-students	Faculty APIs
13	Manage Students	GET	/admin/students	Admin APIs
14	Manage Faculty	GET	/admin/faculty	Admin APIs
15	Approve User Accounts	PUT	/admin/users/approve/{id}	Admin APIs
16	Manage Courses	POST	/admin/courses	Admin APIs
17	Manage Notices	POST	/admin/notices	Admin APIs
18	AI Chatbot	POST	/ai/chat	AI APIs - Gemini Integration
19	Course Recommendation	POST	/ai/recommendation	AI APIs - Gemini Integration
20	Study Plan Generation	POST	/ai/study-plan	AI APIs - Gemini Integration
21	Risk Prediction	POST	/ai/risk-prediction	AI APIs - Gemini Integration
22	Get Notifications	GET	/notifications	Notification APIs
23	Mark Notification as Read	PUT	/notifications/{id}	Notification APIs

Detailed API Specifications

The following endpoint cards describe each API in a readable format so that the content stays within the page boundary and remains easy to review.

A. Authentication APIs

1. User Registration POST /auth/register	
Purpose	Register a new Student or Faculty account in the system.
Input Data	Request body: { "name": "John Doe", "email": "john@example.com", "password": "password123", "role": "student" }
Expected Output	{ "message": "Registration successful. Account pending approval." }
Access Level	Authentication APIs

2. User Login POST /auth/login	
Purpose	Authenticate Student, Faculty, and Admin users using JWT authentication.
Input Data	Request body: { "email": "user@example.com", "password": "password123" }
Expected Output	JWT token and user role. Example: { "token": "jwt_token", "role": "student" }
Access Level	Authentication APIs

3. User Logout POST /auth/logout	
Purpose	Logout an authenticated user from the system.
Input Data	JWT token in authorization header.
Expected Output	Success message. Example: { "message": "Logged out successfully" }
Access Level	Authentication APIs

4. Get User Profile GET /profile	
Purpose	Retrieve profile information of the authenticated user.
Input Data	JWT token in authorization header.
Expected Output	User profile information. Example: { "id": 1, "name": "Student Name", "role": "Student" }
Access Level	Authentication APIs

B. Student APIs

5. View Dashboard GET /student/dashboard	
Purpose	Retrieve the academic performance dashboard of a student.
Input Data	Student ID or authenticated student token.
Expected Output	Attendance percentage, quiz average, and midterm average. Example: { "attendance": "85%", "quiz_average": "78%", "midterm_average": "82%" }
Access Level	Student APIs

6. View Attendance GET /student/attendance	
Purpose	Retrieve attendance records of a student.
Input Data	Student ID or authenticated student token.
Expected Output	Attendance list with dates and attendance status.
Access Level	Student APIs

7. View Academic Results GET /student/results	
Purpose	Retrieve quiz, assignment, and midterm examination results.
Input Data	Student ID or authenticated student token.
Expected Output	Result records including quiz marks, assignment marks, and midterm marks.
Access Level	Student APIs

8. View Risk Alerts GET /student/alerts	
Purpose	Retrieve academic risk alerts generated for the student.
Input Data	Student ID or authenticated student token.
Expected Output	Risk alert information. Example: { "risk_level": "High", "message": "Attendance below threshold" }
Access Level	Student APIs

C. Faculty APIs

9. Monitor Student Performance GET /faculty/students/performance	
Purpose	Allow faculty members to view student performance statistics.
Input Data	Faculty ID or authenticated faculty token.
Expected Output	List of students with academic performance data.
Access Level	Faculty APIs

10. Manage Attendance POST /faculty/attendance	
Purpose	Add or update student attendance records.
Input Data	Request body: { "student_id": 101, "attendance": "Present" }
Expected Output	Success message. Example: { "message": "Attendance updated" }
Access Level	Faculty APIs

11. Manage Assessment Results POST /faculty/results	
Purpose	Upload quiz, assignment, and exam scores of students.
Input Data	Student ID, course ID, assessment type, and marks.
Expected Output	Success message confirming result submission.
Access Level	Faculty APIs

12. View At-Risk Students GET /faculty/risk-students	
Purpose	Identify students with poor academic performance or low attendance.
Input Data	Faculty ID or authenticated faculty token.
Expected Output	List of at-risk students with risk reasons.
Access Level	Faculty APIs

D. Admin APIs

13. Manage Students GET /admin/students	
Purpose	View all student accounts in the system.
Input Data	Admin JWT token.
Expected Output	List of student accounts.
Access Level	Admin APIs

14. Manage Faculty GET /admin/faculty	
Purpose	View all faculty accounts in the system.
Input Data	Admin JWT token.
Expected Output	List of faculty accounts.
Access Level	Admin APIs

15. Approve User Accounts PUT /admin/users/approve/{id}	
Purpose	Approve newly registered user accounts.
Input Data	User ID and admin JWT token.
Expected Output	Success message. Example: { "message": "User approved successfully" }
Access Level	Admin APIs

16. Manage Courses POST /admin/courses	
Purpose	Create and manage course information.
Input Data	Course name, course code, department, and credit.
Expected Output	Success message confirming course creation or update.
Access Level	Admin APIs

17. Manage Notices POST /admin/notices	
Purpose	Publish university notices for users.
Input Data	Notice title, description, date, and target users.
Expected Output	Success message confirming notice publication.
Access Level	Admin APIs

E. AI APIs - Gemini Integration

18. AI Chatbot POST /ai/chat	
Purpose	Allow users to ask academic or university-related questions using Gemini API.
Input Data	Request body: { "question": "What is my exam schedule?" }
Expected Output	AI-generated response. Example: { "response": "Your next exam is on..." }
Access Level	AI APIs

19. Course Recommendation POST /ai/recommendation	
Purpose	Recommend suitable courses based on student academic performance.
Input Data	Student ID and academic records. Example: { "student_id": 101 }
Expected Output	Recommended course list. Example: { "recommended_courses": ["Artificial Intelligence", "Machine Learning"] }
Access Level	AI APIs

20. Study Plan Generation POST /ai/study-plan	
Purpose	Generate a personalized study plan for students.
Input Data	Student academic records, weak subjects, attendance, and performance data.
Expected Output	Generated study plan with suggested study schedule and improvement areas.
Access Level	AI APIs

21. Risk Prediction POST /ai/risk-prediction	
Purpose	Predict academic risk based on attendance and performance data.
Input Data	Attendance percentage, quiz marks, assignment marks, and exam results.
Expected Output	Risk level and reason. Example: { "risk_level": "High", "reason": "Low attendance and poor quiz performance" }
Access Level	AI APIs

F. Notification APIs

22. Get Notifications GET /notifications	
Purpose	Retrieve user notifications, academic alerts, and system messages.
Input Data	User ID or authenticated user token.
Expected Output	Notification list.
Access Level	Notification APIs

23. Mark Notification as Read PUT /notifications/{id}	
Purpose	Update the status of a notification as read.
Input Data	Notification ID and user token.
Expected Output	Success message confirming notification status update.
Access Level	Notification APIs

Summary

The API design follows a modular structure where each user role has separate endpoints. Authentication APIs handle login and logout; Student APIs provide academic information; Faculty APIs manage attendance and assessment results; Admin APIs control users, courses, and notices; AI APIs integrate Gemini-based intelligent services; and Notification APIs handle alerts and system messages.

6

AI Integration Workflow

Google Gemini API Integration — Chatbot, Recommendation, Risk Prediction, Study Plan

6. AI Integration Workflow

6.1 AI Integration Workflow Diagram

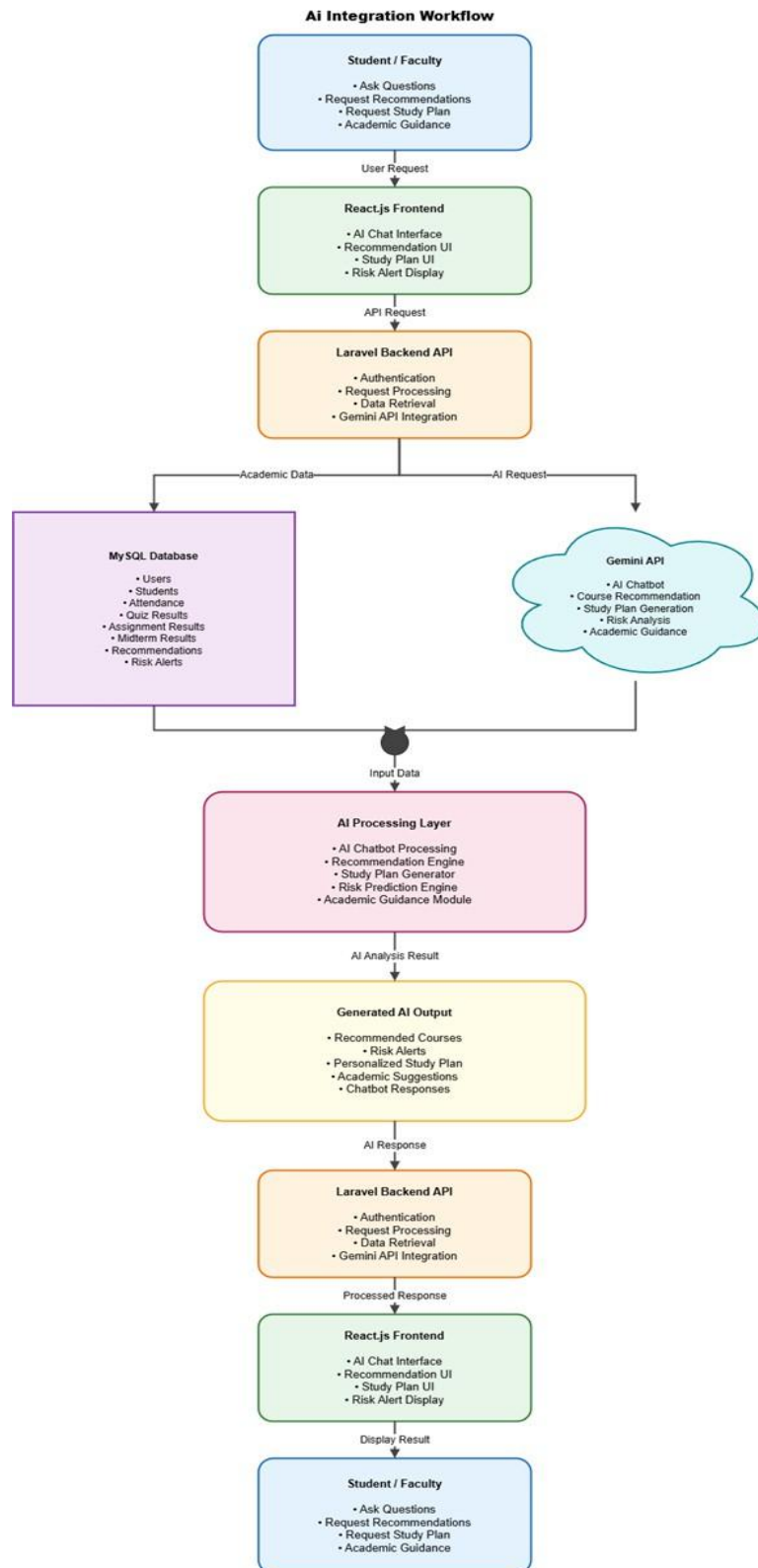


Figure 6.1: AI integration Workflow of AI Smart Campus System

6.2 Why AI is Needed

Artificial Intelligence (AI) is a core component of the Academic Performance Monitoring and AI-Based Recommendation System. The purpose of integrating AI is to provide intelligent academic assistance, personalized learning support, and data-driven decision-making for both students and faculty members.

The AI module analyzes academic performance data, attendance records, assessment results, and user queries to generate meaningful recommendations and insights. Through AI integration, the system can automatically identify at-risk students, suggest suitable courses, create personalized study plans, and provide instant academic guidance through an intelligent chatbot.

6.3 AI Service Used

AI Model: Google Gemini API

The proposed system utilizes Google Gemini API as its primary Artificial Intelligence service. Gemini is integrated with the Laravel Backend API to process academic queries, analyze student performance data, generate recommendations, predict academic risks, and provide intelligent responses.

The Gemini API enables natural language understanding, academic recommendation generation, risk analysis, and personalized learning assistance within the platform.

6.4 Input to the AI System

The AI system receives both academic and user-generated data as input. These inputs are collected from the MySQL Database and user interactions through the React.js Frontend.

Input data include:

- Student attendance records
- Quiz scores and assessment results
- Assignment marks
- Midterm examination results
- Academic performance history
- Course information

- Student and faculty requests
- User questions and academic queries
- Risk-related academic indicators

These inputs are forwarded to the AI service through the Laravel Backend API for further analysis and processing.

6.5 AI Processing Functions

The AI Processing Layer performs several intelligent operations to support academic decision-making and student success.

AI Chatbot

The chatbot answers academic and university-related questions submitted by students and faculty members, including:

- University notices
- Class schedules
- Examination information
- Study-related queries
- Academic assistance requests

Course Recommendation Engine

The recommendation engine analyzes academic records and performance trends to suggest appropriate:

- Core courses
- Elective courses
- Future semester course selections

Study Plan Generator

The study plan generator creates personalized learning schedules and improvement strategies based on a student's academic strengths and weaknesses.

Risk Prediction Engine

The risk prediction module evaluates attendance and academic performance data to identify students who may face academic difficulties due to:

- Low attendance
- Poor assessment performance
- Declining academic progress

Academic Guidance Module

The guidance module provides personalized recommendations and improvement strategies to help students achieve better academic outcomes.

6.6 Output from the AI System

After processing the input data, the AI module generates the following outputs:

- Personalized course recommendations
- Academic risk alerts
- Personalized study plans
- Academic guidance and suggestions
- AI chatbot responses
- Performance improvement recommendations

These outputs are delivered to users through the React.js Frontend interface.

6.7 How AI Improves the Project

The integration of Artificial Intelligence transforms the traditional academic management platform into an intelligent academic support system.

The AI module provides several benefits:

- Personalized learning support for students
- Early detection of academically at-risk students
- Intelligent course recommendation services

- Automated study plan generation
- Instant academic assistance through chatbot interaction
- Enhanced academic monitoring and evaluation
- Improved decision-making support for faculty members
- Better student engagement and learning outcomes

Through these capabilities, AI significantly improves the effectiveness, efficiency, and overall user experience of the system.

6.8 AI Workflow Explanation

Step 1:

A Student or Faculty member submits a question, recommendation request, study plan request, or academic guidance request through the React.js Frontend.

Step 2:

The React.js Frontend sends the request to the Laravel Backend API.

Step 3:

The Backend API authenticates the request and retrieves relevant academic information from the MySQL Database.

Step 4:

The Backend forwards the academic data and user query to the Google Gemini API.

Step 5:

The Gemini AI service analyzes the provided information and performs recommendation generation, study plan creation, chatbot processing, academic guidance, or risk prediction.

Step 6:

The AI-generated result is sent to the AI Processing Layer, where recommendations, alerts, and intelligent outputs are prepared.

Step 7:

The generated AI output is returned to the Laravel Backend API.

Step 8:

The Backend sends the processed response to the React.js Frontend.

Step 9:

The Frontend displays the final result, including recommendations, study plans, risk alerts, academic suggestions, and chatbot responses to the Student or Faculty member.

References

- [1] Google AI for Developers, "Gemini API Documentation." [Online]. Available: <https://ai.google.dev>
- [2] Laravel, "Laravel Framework Official Documentation." [Online]. Available: <https://laravel.com/docs>
- [3] React, "React Official Documentation." [Online]. Available: <https://react.dev>
- [4] Oracle Corporation, "MySQL Documentation." [Online]. Available: <https://dev.mysql.com/doc>
- [5] JWT.io, "JSON Web Token (JWT) Introduction and Documentation." [Online]. Available: <https://jwt.io>
- [6] Draw.io, "Diagram Design Tool." [Online]. Available: <https://app.diagrams.net>
- [7] DBDiagram, "Database Design Tool." [Online]. Available: <https://dbdiagram.io>
- [8] I. Sommerville, *Software Engineering*, 10th ed. Pearson Education, 2016.
- [9] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. McGraw-Hill Education, 2020.